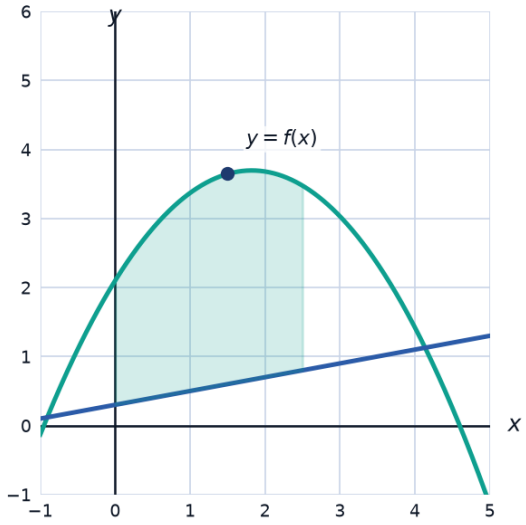
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle -3, 4 \rangle$. What is the vector $2\mathbf{u} + \mathbf{v}$? Show your complete algebraic work.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

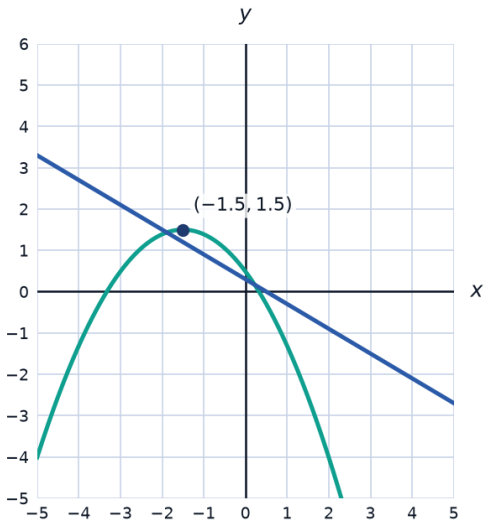
In the complex plane, the point representing the complex number $z_1 = 3 + 4i$ is labeled A. The complex number z_2 is obtained by rotating z_1 counterclockwise by 90° about the origin, and its corresponding point is labeled B. The complex number z_3 is the complex conjugate of z_1, and its corresponding point is labeled C. What is the area of ΔABC? Show your complete algebraic work.



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3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the product of $-4i$ and its complex conjugate.



4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

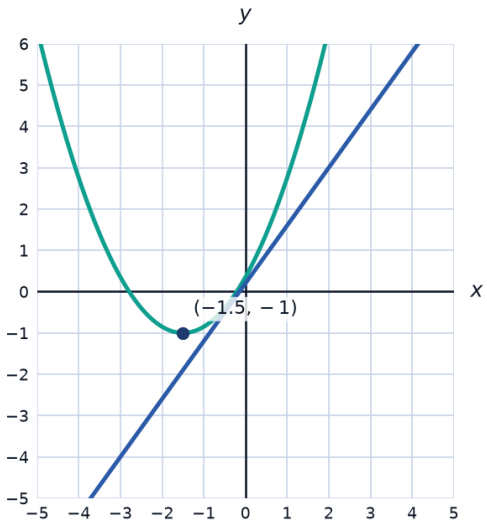
Find the magnitude of the vector $\langle -20, -21 \rangle$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify $(2 + 3i)(1 - i)$ and write in the standard form $a + bi$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A shirt is marked up by 40% and then placed on sale for 25% off. What is the net percentage increase over the original cost?



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Does the ACT Math test penalize wrong guesses, or should every single question be answered before time expires?

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

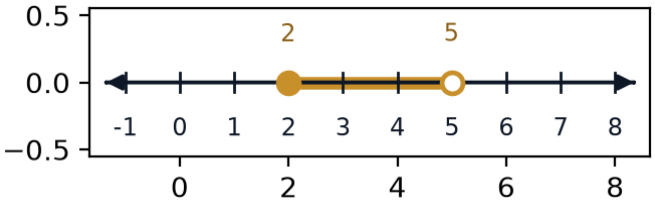
Under the 20-20-20 pacing strategy, how many minutes should remain on the 60-minute clock when you finish Question 20?

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the ordered dataset of 12 numbers: 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 38, 72. (a) Determine Q_1, the median, Q_3, and the IQR. (b) Use Tukey's $1.5 \times \text{IQR}$ rule to determine whether 72 is an outlier.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express the complex number $z = \frac{5+2i}{3-4i}$ in standard form $a + bi$, and determine its modulus $|z|$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a course grading system, Homework is weighted 20%, Midterm Exam is weighted 35%, and Final Exam is weighted 45%. A student scored 92 on Homework and 76 on the Midterm Exam. What is the minimum score the student must achieve on the Final Exam to earn an overall course average of at least 84?

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let matrices $A = (3, -1; 2, 4)$ and $B = (1, 5; 0, -2)$. Compute AB, $\det(A)$, $\det(B)$, and verify that $\det(AB) = \det(A)\det(B)$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the complex expression $\frac{(3-2i)^2}{1+i}$ and write the result in standard form $a + bi$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Section I of EST I Math contains 20 questions in 25 minutes. If a student answers the first 12 questions in an average of 50 seconds each, how many seconds per question are available for the remaining 8 questions?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

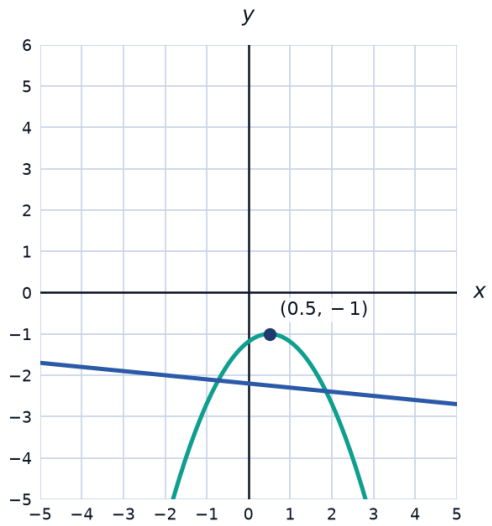
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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

The ACT Subject Test in Mathematics consists of 50 questions in 60 minutes. If a student spends an average of 55 seconds on each of the first 30 questions, how many seconds per question are available for the remaining 20 questions?

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Level 1 · FoundationsThe Digital SAT Math section allocates 70 minutes across two modules of equal length. If each module contains 22 questions, what is the exact average time available per question in seconds?



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Functions & Graphs

Level 4 · StrategyA student finishes the first 18 questions of Module 1 with 9 minutes remaining. Questions 19 and 20 are lengthy geometry proofs, while Questions 21 and 22 are quick Desmos system intersection problems. In what order should the student attack the remaining 4 questions, and why?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

Level 2 · MediumA student solves a linear equation and obtains the fraction $x = -\frac{7}{4}$. What are the valid ways to enter this answer into the Digital SAT Student-Produced Response (SPR) grid?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Plane Geometry

Given vectors $\vec{u} = \langle 3, -4, 2 \rangle$ and $\vec{v} = \langle 2, 1, k \rangle$, find the value of k such that $\vec{u}$ and $\vec{v}$ are orthogonal.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Trigonometry

The mean of 10 numbers is 15. One number is removed and the new mean of the remaining 9 numbers is 14. Find the removed number.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the product of the complex numbers $(4 - 3i)$ and $(2 + 5i)$, where $i^2 = -1$?

a. $23 + 14i$

b. $7 + 14i$

c. $23 - 14i$

d. $8 - 15i$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the simplified form of the quotient $\frac{5 - i}{2 + i}$?

a. $\frac{9 - 7i}{5}$

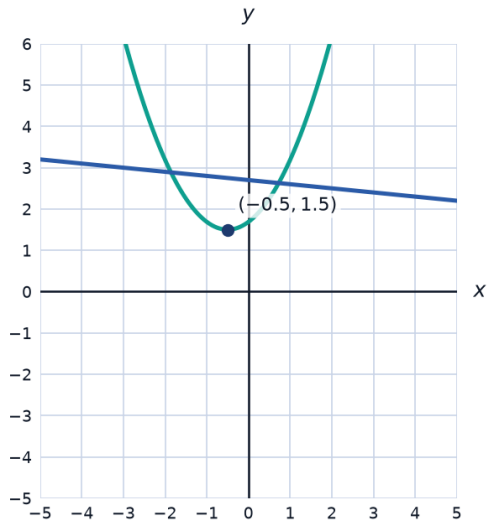
b. $\frac{11 - 7i}{5}$

c. $3 - i$

d. $\frac{9 + 3i}{5}$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the value of i^{47}, where $i = \sqrt{-1}$?



a. 1

b. -1

c. i

d. $-i$

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the modulus (absolute value) of the complex number $z = -5 + 12i$?

a. 7

b. 13

c. 17

d. $\sqrt{119}$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the distance in the complex plane between $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$?

a. $2\sqrt{13}$

b. $\sqrt{20}$

c. 10

d. 5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $z = 2 + 2i$, what is the argument θ of z in radians, where $0 \le \theta < 2\pi$?

a. $\frac{\pi}{6}$

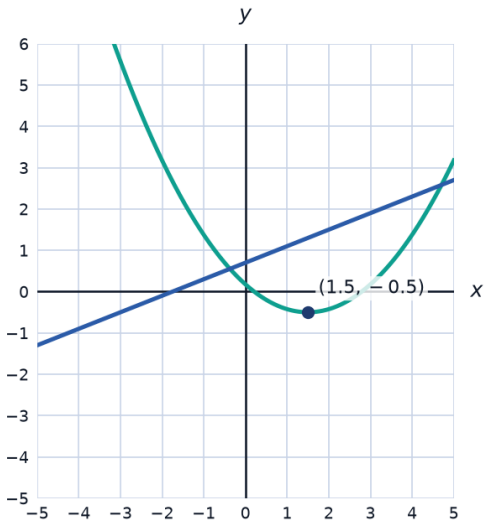
b. $\frac{\pi}{4}$

c. $\frac{\pi}{3}$

d. $\frac{3\pi}{4}$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the determinant of the 2×2 matrix $M = [[4, 6], [-2, 3]]$?



a. 0

b. 12

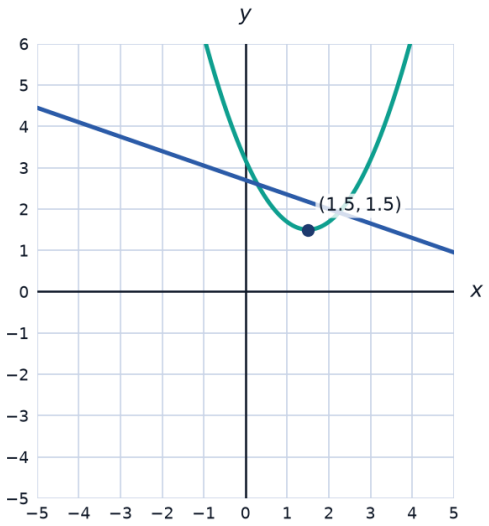
c. 24

d. 36

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of k will the matrix $K = [[k, 6], [2, 3]]$ NOT have an inverse?



a. 1

b. 2

c. 3

d. 4

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$ are given. What are the components of $2\mathbf{u} + \mathbf{v}$?

a. $\langle 4, -3 \rangle$

b. $\langle 1, 1 \rangle$

c. $\langle 4, 3 \rangle$

d. $\langle -1, -3 \rangle$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the dot product of the vectors $\mathbf{a} = \langle 5, -2 \rangle$ and $\mathbf{b} = \langle 4, 7 \rangle$?

a. 6

b. 26

c. -6

d. 34

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the magnitude of the vector $\mathbf{w} = \langle -9, 12 \rangle$?

a. 13

b. 15

c. 17

d. 21

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the complex number system, for what real value of k is the expression $\frac{k+8i}{2-5i}$ a purely imaginary number? (Note: $i = \sqrt{-1}$)

a. -20

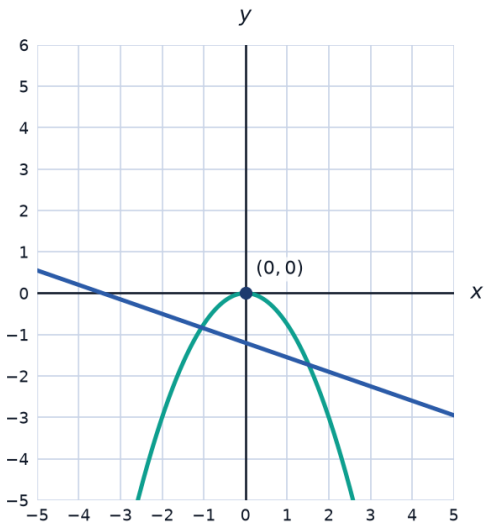
b. $-\frac{16}{5}$

c. $\frac{16}{5}$

d. 20

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the imaginary unit i, where $i^2 = -1$, which of the following complex numbers is equal to $(5 - 3i) - (2 - 8i)$?



a. $3 + 5i$

b. $3 - 11i$

c. $7 - 11i$

d. $3 - 5i$

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

If $A = (2, -11; 9, -3)$ and $B = (-14, -11; 11, -5)$, find the entry in row 1, column 2 of $A + B$.

a. -25

b. -21

c. -16

d. -22

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the complex conjugate of $11 - 7i$.

a. 11-7i

b. 11+7i

c. -11+7i

d. -11-7i

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the determinant of $(-5, -7; -3, -7)$.

a. 9

b. 18

c. 16

d. 14

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Write 1×10^4 in standard (decimal) form.

a. 20000

b. 1000

c. 10000

d. 100000

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

A quantity increases from 15 to 18. What is the percent increase?

a. 20\%

b. 16\%

c. 29\%

d. 10\%

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Plane Geometry

If 6 notebooks cost $x24, what is the cost per notebook?

a. $x23

b. $x33

c. $x1

d. $x4

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Trigonometry

A quantity decreases from 10 to 5. What is the percent decrease?

a. 30\%

b. 20\%

c. 24\%

d. 50\%

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

If 2 tickets cost $x12$, what is the cost per ticket?

a. $x6$

b. $x7$

c. $x8$

d. $x17$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

If 3 pencils cost $x9$, what is the cost per pencil?

a. $x16$

b. $x12$

c. $x6$

d. $x3$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Given $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 3 \rangle$, what is the cross product $\mathbf{u} \times \mathbf{v}$?

a. $\langle 2, 3, 0 \rangle$

b. $\langle 6, 3, 1 \rangle$

c. $\langle 6, -3, 1 \rangle$

d. $\langle -6, -3, -1 \rangle$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

In the standard (x, y) coordinate plane, a laser beam emitted from point $A(-3, 8)$ reflects off the x-axis at point $P(c, 0)$ and travels directly to target $B(5, 4)$. Given that the angle of incidence equals the angle of reflection, what is the value of c?

a. 1

b. $\frac{5}{3}$

c. $\frac{7}{3}$

d. $\frac{8}{3}$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Coordinate Geometry

If $\sin\theta = \frac{7}{25}$ and θ is in Quadrant I, find $\cos\theta$.

a. $-\frac{7}{25}$

b. $\frac{24}{25}$

c. $-\frac{24}{25}$

d. $\frac{7}{25}$

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